



RAOH

— R e s e a r c h e r s —

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

Methodology

Purpose of the Methodology

The Methodology explains exactly how the study was conducted so that others can understand, evaluate, and potentially replicate it.

Main Components

1. Study Design

State the type of study.

Examples:

- Cross-sectional study
- Case-control study
- Cohort study
- Randomized controlled trial
- Systematic review
- Narrative review

2. Study Setting and Population

Describe where and when the study was conducted and who participated.

Example:

This cross-sectional study was conducted among medical students at Yarmouk University between January and March 2026.

3. Inclusion and Exclusion Criteria

Clearly define who was eligible and who was excluded.

4. Sample Size and Sampling Method

Explain how participants were selected and how the sample size was determined.

Examples:

- Convenience sampling
- Random sampling
- Stratified sampling

5. Data Collection

Describe the tools and procedures used to collect data.

Examples:

- Online questionnaire

- Medical records
- Laboratory measurements
- Interviews

6. Variables

Define the main variables.

Examples:

- Independent variable
- Dependent variable
- Confounding variables

7. Statistical Analysis

Describe how the data were analyzed.

Examples:

- Mean $\pm$ SD
- Frequencies and percentages
- Chi-square test
- Independent t-test
- ANOVA
- Logistic regression
- Pearson correlation

Specify the software used (e.g., SPSS, R, Jamovi, Stata).

8. Ethical Considerations

State ethical approval and participant consent procedures.

Example:

Ethical approval was obtained from the Institutional Review Board (IRB), and informed consent was obtained from all participants.

Writing Tips

- Write in past tense.
- Be specific and reproducible.
- Describe what was actually done.
- Include enough detail for replication.
- Use subheadings for clarity.

Citation Tips

- Cite validated questionnaires and measurement tools.
- Cite diagnostic criteria, guidelines, or scoring systems.
- Statistical tests generally do not require citations unless a specialized method is used.
- Cite the original source when using a published instrument.

Example

Correct:

Sleep quality was assessed using the Pittsburgh Sleep Quality Index (PSQI).¹

Incorrect:

Sleep quality was assessed using a questionnaire.

(The questionnaire is not identified or referenced.)

Common Mistakes

- Missing inclusion/exclusion criteria.
- Insufficient detail about data collection.
- Not reporting sampling methods.
- Omitting ethical approval information.
- Describing results in the Methodology section.
- Using future tense instead of past tense.

AI Usage

AI may assist with language editing and manuscript writing if permitted by the journal and appropriately disclosed. However, researchers must independently perform, verify, and interpret all analyses, results, and references.

Final Checklist

- ✓ Study design clearly stated.
- ✓ Population and setting described.
- ✓ Inclusion and exclusion criteria provided.
- ✓ Sampling method explained.
- ✓ Data collection procedures described.
- ✓ Statistical analysis specified.
- ✓ Ethical approval reported.
- ✓ Sufficient detail provided for replication.

Golden Rule

A reader should be able to understand exactly what you did, who you studied, how you collected the data, and how you analyzed it.